Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [14MES2]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2018

TURBO MACHINERY

Time:3 Hours Maximum Marks:70

Instructions:
i. Answer any **FIVE** full Questions choosing atleast two from each part.
ii. Missing data may be suitably assumed.
iii. Use of steam stables and data hand book/mollier chart is permitted

PART - A

1. a. Define turbomachine. Explain the parts of a turbomachine with a neat sketch. 06

   b. The efficiency of a fan (η) depends upon density ρ, viscosity μ, angular velocity ω, diameter of the rotor D and the discharge Q. Obtain an expression for η in terms of dimensionless parameters. 08

2. a. Derive an alternate form of Euler's turbine equation and explain the various components of energy transfer. 06

   b. In a mixed flow turbomachine, fluid enters with an absolute velocity axially at inlet and leaves with a relative velocity radially at outlet. Calculate the degree of reaction and energy input to the machine if the relative velocity at outlet is equal to the tangential blade speed at inlet. Take the following data: Inlet diameter=8cm, exit diameter=25 cm, speed=3000 rpm, blade angle at inlet=45^0. 08

3. a. Derive an expression for the theoretical head-capacity relationship of a centrifugal pump. 06

   b. Draw the velocity triangles at inlet and outlet of an axial flow compressor with the following data: degree of reaction (R) = 0.5, Inlet blade angle with respect to axial direction = 45^0, constant axial flow velocity =120 m/s, radius of rotation = 0.2 m, compressor speed = 6500 rpm. Determine the power required to handle 15 kg of air/second. 08

4. a. Explain the various heads and efficiencies applicable to hydraulic turbines. 06

   b. A 3-jet Pelton wheel is required to generate 10,000 kW of power under a net head of 400 m. The blade angle at outlet is 15^0 and the reduction in the velocity (relative) while passing over the blade is 5%. If the overall efficiency of the wheel is 80%, C_v = 0.98 and speed ratio = 0.46, determine 08

   i. the diameter of the jet
   ii. the total flow rate
   iii. force exerted by the jet on the buckets.

PART - B

5. a. Explain the various types of draft tube with neat sketches. 06

b. A Kaplan turbine develops 1500 kW of power under a head of 6 m. The turbine is set at 2.5 m above the tailrace level. A vacuum gauge inserted at the turbine outlet shows a suction head of 3.2 m. If the turbine efficiency is assumed to be 85%, calculate the draft tube efficiency. Take the inlet diameter of draft tube as 3 m. 08

6. a. Explain, with schematic sketches, velocity compounding and pressure compounding of a single stage steam turbine. Draw the pressure and velocity diagrams for the two types of compounding. 06

b. In a reaction turbine the blade speed is 65 m/s. Dry steam at 1.5 bar flows at 5 kg/s. The blade angles are 20^0 and 35^0. Calculate 08

i. The blade height which is $1/10^{th}$ of the blade ring diameter

ii. Power developed

iii. Enthalpy drop if the stage efficiency is 80%.

7. a. Explain with relevant expressions the following with respect to centrifugal pump: 06

i. Suction head

ii. Delivery head

iii. Total head

iv. Net positive suction head.

b. A 4-stage centrifugal pump supplying water is to be designed for a total lift of 120 m, while running at 1450 rpm. The discharge is 0.24 m^3/s. The vanes are set back at 30^0 with the wheel tangent at outlet and impeller is surrounded by guide vanes. Water enters the vane passages in a radial direction and the flow velocity through the impeller is 0.3 times the outlet peripheral velocity. Determine the diameter and width of the impeller at outlet, the manometric efficiency and the angle of the guide vane. 08

8. a. Derive an expression for the overall pressure ratio (stagnation state) developed by a centrifugal compressor. 06

b. An axial flow compressor has the following data: Entry condition-1 bar, 20^0C, Degree of reaction – 50%, Mean blade ring diameter = 36 cm, Rotational speed = 18,000 rpm, Blade height at entry = 6 cm, Blade angle at rotor and stator exit = 65^0, Axial velocity = 180 m/s, Mechanical efficiency = 96.7%. Calculate 08

i. The blade angles at rotor and stator inlet

ii. Power required to drive the compressor